

Ex No: 1d

IMPLEMENTATION OF A* SEARCH ALGORITHM

Aim:

To implement A* Search Algorithm

Procedure:

Define a Node class storing state, parent, path cost (g), heuristic cost (h), and total cost ($f = g + h$).

Initialize a_star_search using a priority queue (open_list) for unexplored nodes and a set (closed_list) for visited states.

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Pop the lowest-f node, reconstruct and return the path if the goal is reached, or add it to closed_list.

Evaluate adjacent neighbors, calculating g, h, and total f, pushing valid new neighbors onto open_list.

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Define Manhattan distance heuristic and grid neighbors helper functions, then run a_star_search(start, goal) to find and print the shortest path.

Program:

```
import heapq

class Node:

    def init (self, state, parent, g, h):

        self.state = state
        self.parent = parent
        self.g = g # Cost to reach the current node
        self.h = h # Heuristic cost from current node to goal
        self.f = g + h # Total cost (f = g + h)

    def lt (self, other):

        return self.f < other.f

def a_star_search(start, goal, heuristic, neighbors):

    open_list = []
    closed_list = set()

    start_node = Node(start, None, 0, heuristic(start, goal))
    heapq.heappush(open_list, start_node)
```

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while open_list:
    current_node = heapq.heappop(open_list)

    if current_node.state == goal:
        # Reconstruct the path from start to goal
        path = []
        while current_node:
            path.append(current_node.state)
            current_node = current_node.parent
        return path[::-1]

    closed_list.add(current_node.state)

    for neighbor, cost in neighbors(current_node.state):
        if neighbor in closed_list:
            continue

        g = current_node.g + cost
        h = heuristic(neighbor, goal)
        neighbor_node = Node(neighbor, current_node, g, h)

        if not any(open_node.state == neighbor and open_node.f <= neighbor_node.f for
open_node in open_list):
            heapq.heappush(open_list, neighbor_node)

    return None # If no path is found
# Example of usage

def heuristic(state, goal):
    # Simple Manhattan distance heuristic for a grid (assuming coordinates)
    return abs(state[0] - goal[0]) + abs(state[1] - goal[1])

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```
def neighbors(state):
    # Define neighbors as adjacent grid cells (up, down, left, right)
    x, y = state
    return [
        ((x + 1, y), 1), # Right
        ((x - 1, y), 1), # Left
        ((x, y + 1), 1), # Up
        ((x, y - 1), 1) # Down
    ]

start = (0, 0)
goal = (3, 3)

path = a_star_search(start, goal, heuristic, neighbors)
print("Path from start to goal:", path)
```

Output:

For start = (0, 0) and goal = (3, 3):

Path from start to goal: [(0, 0), (1, 0), (2, 0), (3, 0), (3, 1), (3, 2), (3, 3)]